1. A Box contains 6 red Balls and 4 green balls. Two balls are drawn at random, one after the other, without replacement. What is the probability that the second ball drawn is green, given that the first ball drawn was red?

Soln:

Given that,

A box contains 6 red balls and 4 green balls, total 10 balls two balls and drawn without replacement.

Given that the first ball is red

Since the first ball is red, one red ball is removed

Now box contains 9 balls

let P(E) be the probability of getting second ball drawn is given

$$P(E) = 4/9 = \frac{\text{Number of green balls}}{\text{total balls.}}$$

2) A factory produces 3 types of machines. Type A, Type B and Type C The probability of producing each type of machine is as follows $P(A) = 0.5$ $P(B) = 0.2$ $P(C) = 0.2$ the probability of a machine being defective given its type is $P(D/A) = 0.02$, $P(D/B) = 0.05$, $P(D/c) = 0.01$ A machine is selected at random and it is found to be defective. What is the probability that the defective is of type A?

Soln:

Given that,

$P(A) = 0.5$

$P(B) = 0.3$

$P(C) = 0.2$

Defective probabilities,

$P(D/A) = 0.02$

$P(D/B) = 0.05$

$P(D/C) = 0.01$

According to Baye's Theorem

We need to find the probability that a defective machine is Type A

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$$P(A/D) = \frac{P(D/A) \cdot P(A)}{P(D)}$$

$$P(D) = P(D/A) \cdot P(A) + P(D/B) \cdot P(B) + P(D/C) \cdot P(C)$$

$$= 0.027$$

$$\therefore P(A/D) = \frac{0.02 \times 0.5}{0.027} = \frac{0.01}{0.027} = \frac{10}{27} = 0.3703$$

3) A Bag A contains 2 white and 3 red balls and bag B contains 4 white and 5 red balls one ball is drawn at random from one of the bags and it is found to be red. Find the probability that the red ball drawn is from bag B.

Sol:- Given,

Bag A contains 2 white and 3 red balls

$$P(R/A) = \frac{3}{5} = 0.6$$

Bag B contains 4 white and 5 red balls

$$P(R/B) = 5/9$$

Probability of getting Bag A

$$P(A) = 1/2$$

Probability of getting Bag B

$$P(B) = 1/2$$

The probability that the red ball drawn is from Bag B

$$P(B/R) = \frac{P(R/B) \cdot P(B)}{P(R)}$$

$$P(R) = P(R/A)\ P(A) + P(R/B) \cdot P(B)$$

$$= \frac{3}{5} \times \frac{1}{2} + \frac{5}{9} \times \frac{1}{2}$$

$$= \frac{27}{90} + \frac{25}{90}$$

$$= \frac{52}{90} = \frac{26}{45}$$

$$P(B/R) = \frac{\frac{5}{9} \times \frac{1}{2}}{\frac{26}{45}} = \frac{5}{18} \times \frac{45}{26}$$

$$P(B/R) = \frac{25}{52}$$

$$= 0.4808$$

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4) A Business man goes to hotels X, Y, Z, 20%, 50% & 30% of the time respectively. It is shown that 5%, 4% & 8% of the rooms in X, Y, Z hotels have faculty plumbing. What is the probability that business man's room having faculty Plumbing is assigned to hotal X ?

Sol:- Let X, Y, Z denotes the events of a business man going to hotels X, Y, Z respectively.

Also, let 'F' be denote the events of a hotel room to have faculty plumbings.

Given, $P(X) = 20\% = 0.2$

$P(Y) = 50\% = 0.5$

$P(Z) = 30\% = 0.3$

Also given, $P(F/X) = 5\% = 0.5$

$P(F/Y) = 4\% = 0.04$

$P(F/Z) = 8\% = 0.08$

Required Probability is $P(X/F) = \frac{P(F/X) \cdot P(X)}{P(F)}$

$P(F) = P(F/X) \cdot P(X) + P(F/Y) P(Y) + P(F/Z) \cdot P(Z)$

$= 0.054$

$P(X/F) = \frac{0.05 \times 2}{0.054} = \frac{5}{27} = 0.1851$

5) Suppose 5 man out of 100 and 25 woman out of 10,000 are colour blind. A colour blind person is choosen at random. What is the Probability that the person being male

Sol:- Given,

Probability that a person is male:

$P(M) = 1/2$

probability that a Person is female:

$P(F) = 1/2$

Probability that a male is colour blind:

$P(C/M) = \frac{5}{100} = 0.05$

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Probability that a female is colour blind:

$$P(C/F) = \frac{25}{10.000} = 0.0025$$

The Probability that a colour blind person is male.

$$P(M/C) = \frac{P(C/M)\,P(M)}{P(C)}$$

$$P(C) = P(C/M)\,P(M) + P(C/F).\,P(F)$$

$$= 0.02625$$

$$P(M/C) = \frac{0.025}{0.02625} = \frac{20}{21} = 0.9524$$

6Y. A random variable X has the following probability function

X	0	1	2	3	4	5	6	7
P(X)	0	K	2K	2K	3K	K^2	$2K^2$	$7K^2+K$

Determine, (i) K (ii) $P(X<6)$, $P(X\geq 6)$ & $P(0<X<5)$

(iii) If $P(X\leq k) > 1/2$, find the maximum value of 'k'

(iv) Standard deviation of X and

(v) Cummulative distribution of X

Sol: (i) We know that Sum of probabilities = 1

$$K+2K+3K+2K+K^2+2K^2+7K^2+K=1$$

$$9K+10K^2=1$$

$$10K^2+9K-1=0$$

$$10K^2+10K-K-1=0$$

$$10K(K+1)-1(K+1)=0$$

$$(10K-1)(K+1)=0$$

$$K=1/10 \quad K=-1$$

As K must be greater than 0, we take $K=\frac{1}{10}$, $= 0.1$

(ii) a) $P(X<6) = P(X=0)+P(X=1)+P(X=2)+P(X=3)+P(X=4)+P(X=5)$

$$= K+2K+2K+3K+K^2$$

$$= 8K+K^2$$

$$= 8(0.1)+(0.1)^2$$

$$= 0.81$$

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(b) $P(X \geq 6) = P(6) + P(7)$

$= 2(0.1)^2 + 7(0.1)^2 + 0.1$

$= 0.17$

(c) $P(0 < X < 5) = P(1) + P(2) + P(3) + P(4)$

$= 0.1 + 0.2 + 0.2 + 0.3$

$= 0.8$

(iii) Find the maximum value of k such that

$P(X \leq k) > 1/2$

for $k=0$, $P(X \leq 0) = 0 \not> 1/2$

$k=1$, $P(X \leq 1) = 0.1 \not> 1/2$

$k=2$, $P(X \leq 2) = P(X=0) + P(X=1) + P(X=2)$

$= 0.3 \not> 1/2$

$k=3$, $P(X \leq 3) = P(X=0) + P(X=1) + P(X=2) + P(X=3)$

$= 0.5 \not> 1/2$

$k=4$, $P(X \leq 4) = 0.8 > 1/2$

$\therefore$ Minimum value of k is $k=4$

(iv) Standard deviation

Mean of $X = \sum_{i=1}^{8} x P(x)$

$= 1(0.1) + 2(0.2) + 3(0.2) + 4(0.3) + 5(0.01) + 6(0.02) + 7(0.17)$

$= 3.66$

$E(X^2) = \sum x^2 P(X)$

$= 1^2(0.1) + 2^2(0.2) + 3^2(0.2) + 4^2(0.3) + 5^2(0.01) + 6^2(0.02) + 7^2(0.17)$

$= 16.80$

Variance, $\sigma^2 = E(X^2) - [E(X)]^2$

$= 16.80 - (3.66)^2$

$= 3.4044$

$\sigma = \sqrt{3.4044} = 1.845$

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(v) Cummulative distribution function of X is $F(X) = P(X \le x)$

Now $F(x) = 0$, $x < 0$

$F(0) = P(X \le 0) = 0$, $0 \le x < 1$

$F(1) = P(X \le 1) = k = 0.1$, $1 \le x < 2$

$F(2) = P(X \le 2) = 3k = 0.3$, $2 \le x < 3$

$F(3) = P(X \le 3) = 5k = 0.5$, $3 \le x < 4$

$F(4) = P(X \le 4) = 8k = 0.8$, $4 \le x < 5$

$F(5) = P(X \le 5) = 8k + k^2 = 0.81$, $5 \le x < 6$

$F(6) = P(X \le 6) = 8k + 3k^2 = 0.83$, $6 \le x < 7$

$F(7) = P(X \le 7) = 10k^2 + 9k = 1$, $x \ge 7$

7) Two dice are known let X assign to each point (a,b) in S the maximum of its numbers i.e., $X(a,b) = \max(a,b)$. Find the Probability distribution. X is a random variable with $X(S) = \{1,2,3,4,5,6\}$. Also find the mean and variance of its distribution.

Sol:- Given the pair of dice is thrown

its Sample size, $S = \{ (1,1), (1,2) \ldots (1,6), \vdots (6,1), (6,2) \ldots (6,6) \}$

$\rightarrow n(S) = 36$

Given X assign to each Point (a,b) in S the maximum of its numbers i.e., $X(a,b) = \max$

$\Rightarrow X = \{1,2,3,4,5,6\}$

Now, $P(X=1) = P_1 = 1/36$

$P(X=2) = P_2 = 3/36$

$P(X=3) = P_3 = 5/36$

$P(X=4) = P_4 = 7/36$

$P(X=5) = P_5 = 9/36$

$P(X=6) = P_6 = 11/36$

(i) Probability distxibution of X:-

$x_i = X$	1	2	3	4	5	6
$P(X = x_i)$	1/36	3/36	5/36	7/36	9/36	11/36

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(ii) Mean of X is

$$\mu = \sum_{i=1}^{6} x_i P(X = x_i)$$

$$\mu = \left[1 \times \frac{1}{36}\right] + \left[2 \times \frac{3}{36}\right] + \left[3 \times \frac{5}{36}\right] + \left[4 \times \frac{7}{36}\right] + \left[5 \times \frac{9}{36}\right] + \left[6 \times \frac{11}{36}\right]$$

$$= \frac{1+6+15+28+45+66}{36}$$

$$= 4.47 = E(X)$$

(iii) Variance of X is given by

$$\sigma^2 = V(X) = E(X^2) - [E(X)]^2$$

$$E(X^2) = \sum_{i=1}^{6} x_i^2 P(X = x_i)$$

$$= \left[1^2 \times \frac{1}{36}\right] + \left[2^2 \times \frac{3}{36}\right] + \left[3^2 \times \frac{5}{36}\right] + \left[4^2 \times \frac{7}{36}\right] + \left[5^2 \times \frac{9}{36}\right] + \left[6^2 \times \frac{11}{36}\right]$$

$$= \frac{1}{36} + \frac{12}{36} + \frac{45}{36} + \frac{112}{36} + \frac{225}{36} + \frac{396}{36} = \frac{791}{36}$$

$$E(X^2) = 21.97$$

$$V(X) = E(X^2) - [E(X)]^2 = 21.97 - (4.47)^2$$

$$= 21.97 - 19.98$$

$$= 1.99$$

8) For the following Probability distribution.

X	-3	6	9
P(X)	1/6	1/2	1/3

Find i) E(X) ii) $E(X^2)$ iii) $E(2X+1)^2$

Sol:- (i) Mean of X is

$$\mu = \sum_{i=1} x_i [P(X = x_i)]$$

$$= (-3)(1/6) + 6(1/2) + 9(1/3)$$

$$= 5.5 = E(X)$$

(ii) $E(X^2) = \sum_{i=1}^{3} x_i^2 P(X = x_i)$

$$= (-3)^2 (1/6) + (6)^2 (1/2) + (9)^2 (1/3)$$

$$= \frac{9}{6} + \frac{36}{2} + \frac{81}{3}$$

$$= 46.5$$

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(iii) $E(2X+1)^2 = E(4X^2+1+4X)$

$= 4E(X^2)+4E(X)+1$

$= 4(46.5)+4(5.5)+1$

$= 209$

9) The Probability density function of a Cotinous random Variable is given by

$$f(x) = \begin{cases} K(1-x^2) & , 0<x<1 \\ 0 & , \text{otherwise} \end{cases}$$

(i) Find K (ii) Probability between 0.1 & 0.2

(iii) Standard deviation.

Soln: (i) $\int_0^1 K(1-x^2)\,dx = 1$

$K\left[x - \frac{x^3}{3}\right]_0^1 = 1$

$K\left[1-\frac{1}{3}\right] = 1$

$K\left(\frac{2}{3}\right) = 1$

$\boxed{K = 3/2}$

(ii) $P(0.1<x<0.2) = \int_{0.1}^{0.2} \frac{3}{2}(1-x)^2 = \frac{3}{2}\left[x - \frac{x^3}{3}\right]_{0.1}^{0.2}$

$= \frac{3}{2}\left[0.2 - \frac{0.008}{3} - \left[0.1 - \frac{0.001}{3}\right]\right]$

$= \frac{3}{2}\left[0.197333 - 0.099667\right]$

$= 0.146499$

(iii) Standard deviation

$E(X) = \int_0^1 x \cdot \frac{3}{2}(1-x^2)\,dx$

$= \frac{3}{2}\int_0^1 (x-x^3)\,dx$

$= \frac{3}{2}\left[\frac{x^2}{2} - \frac{x^4}{4}\right]_0^1 = \frac{3}{2}\left[\frac{1}{2}-\frac{1}{4}\right] = \frac{3}{2}\cdot\frac{1}{4} = 3/8$

$E(X^2) = \frac{3}{2}\int_0^1 (x^2-x^4)\,dx$

$= \frac{3}{2}\left[\frac{x^3}{3} - \frac{x^5}{5}\right]_0^1 = \frac{3}{2}\left[\frac{1}{2}-\frac{1}{5}\right] = \frac{3}{2}\left(\frac{2}{15}\right) = 1/5$

$\sigma^2 = E(X^2) - (E(X))^2$

$= \frac{1}{5} - \left(\frac{3}{8}\right)^2 = \frac{1}{5} - \frac{9}{64} = \frac{19}{320} = 0.2437$

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10) A continous random variable X has the following probability distribution function $f(x) = kx^2e^{-x}$, $x \geq 0$ find k, mean & variance.

Sol:-

$$F(x) = \begin{cases} kx^2e^{-x}, & x \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

$$\int_0^\infty kx^2e^{-x}\,dx = 1$$

$$k\left[-x^2e^{-x} - \int 2x\,(-e^{-x})\right]_0^\infty = 1$$

$$= k\left[-x^2e^{-x} + 2\left(x(-e^{-x}) - \int(-e^{-x})\,dx\right)\right]_0^\infty = 1$$

$$= k\left[-x^2e^{-x} + 2(x(-e^{-x}) + (-e^{-x}))\right]_0^\infty = 1$$

$$= k\left[-e^{-x}(x^2+2x+2)\right]_0^\infty = 1$$

$$= k\left[-e^{-\infty}(\infty^2+2\infty+2) + 2^\circ(0+0+2)\right] = 1$$

$$k[0+2] = 1$$

$$k(2) = 1$$

$$\boxed{k = 1/2}$$

(ii) Mean

$$E(X) = \int_0^\infty x f(x)\,dx$$

$$= \frac{1}{2}\int_0^\infty x^3e^{-x}\,dx = \frac{1}{2}\left[-x^3e^{-x} + 3\int x^2e^{-x}\,dx\right]_0^\infty$$

$$= \frac{1}{2}\left[-x^3e^{-x} + 3\left(x^2(-e^{-x}) + \int 2xe^{-x}\,dx\right)\right]_0^\infty$$

$$E(X) = \frac{1}{2} \times 6 = 3$$

(iii) Variance

$$E(X^2) = \int_0^\infty x^2 f(x)\,dx$$

$$= \frac{1}{2}\int_0^\infty x^4e^{-x}\,dx$$

$$= \frac{1}{2}\left[-x^4e^{-x} + 4\int x^3e^{-x}\,dx\right]_0^\infty$$

$$= \frac{1}{2} \times 24 = 12$$

$$Var(X) = E(X^2) - [E(X)]^2$$

$$= 12 - 9$$

$$\sigma^2 = 3$$

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11) A continous random variable X has the following cumulative distribution function

$$F(x) = \begin{cases} 0 & \text{if } x \le 1 \\ k(x-1)^3 & \text{if } 1 \le x \le 3 \\ 1 & \text{if } x > 3 \end{cases}$$

Find (i) k, (ii) Probability distribution function

(iii) Mean

Sol:- (i) Given $F(3) = 1$

$$k(3-1)^3 = 1$$

$$8k = 1$$

$$\boxed{k = 1/8}$$

(ii) $f(x) = \frac{d}{dx} F(x)$

For $1 < x < 3$

$$f(x) = \frac{d}{dx}\left[\frac{1}{8}(x-1)^3\right]$$

$$= \frac{1}{8} 3(x-1)^2$$

$$f(x) = \frac{3}{8}(x-1)^2$$

$$f(x) = \begin{cases} 0 & x \le 1 \\ \frac{3}{8}(x-1)^2 & 1 < x < 3 \\ 0 & x \ge 3 \end{cases}$$

(iii) Mean

$$E(X) = \int_{-\infty}^{\infty} x f(x)\, dx$$

$$= \frac{3}{8}\int_1^3 x(x-1)^2 dx$$

$$x(x-1)^2 = x(x^2 - 2x + 1) = x^3 - 2x^2 + x$$

$$E(x) = \frac{3}{8}\int_1^3 (x^3 - 2x^2 + x)\, dx$$

$$= \frac{3}{8}\left[\frac{x^4}{4} - \frac{2x^3}{3} + \frac{x^2}{2}\right]_1^3$$

$$= \frac{3}{8}\left[\frac{27}{4} - \frac{1}{12}\right]$$

$$= \frac{3}{8}\left(\frac{80}{12}\right)$$

$$E(X) = \frac{5}{2}$$

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12) let the random variable X have the p.d.f

$$f(x) = \begin{cases} \frac{1}{2} e^{-x/2}, & x > 0 \\ 0, & \text{otherwise} \end{cases}$$

Find the moment generating function, mean & variance of X

Sol:- Given,

$$f(x) = \begin{cases} \frac{1}{2} e^{-x/2}, & x > 0 \\ 0, & \text{otherwise} \end{cases}$$

(i) Moment Generating function

$$M_X(t) = E(e^{tx})$$

$$= \int_0^\infty e^{tx} \cdot \frac{1}{2} e^{-x/2} dx = \frac{1}{2} \int_0^\infty e^{(t-1/2)x} dx$$

$$\int_0^\infty e^{-ax} dx = \frac{1}{a}, \quad a > 0 \qquad t < 1/2$$

$$a = \frac{1}{2} - t$$

$$M_X(t) = \frac{1}{2} \cdot \frac{1}{\frac{1}{2} - t}$$

$$M_X(t) = \frac{1}{1-2t}, \quad t < 1/2$$

(ii) Mean

$$M'_X(t) = \frac{2}{(1-2t)^2}$$

$$E(X) = M'_X(0) = 2$$

(iii) Variance

$$M''_X(t) = \frac{8}{(1-2t)^3}$$

$$E(X^2) = M''_X(0)$$

$$= 8$$

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